



ok how about with these results tell me:

Silhouette Score: 0.4239

Davies-Bouldin Score: 1.6507

Calinski-Harabasz Score: 122.6928

Based on your results, here's how to interpret them:

Result Interpretation

Metric	Your Value	Interpretation
Silhouette Score	0.4239	⚖️ Moderate clustering quality ^[1] ^[2]
Davies-Bouldin Score	1.6507	⚠️ Moderate-poor (lower is better; ideal < 1) ^[1] ^[2]
Calinski-Harabasz Score	122.6928	⚖️ Moderate (higher is better; depends on dataset size) ^[1] ^[2]

Silhouette Score: 0.4239

Range	Quality
0.8 – 1.0	Excellent
0.5 – 0.8	Good
0.25 – 0.5	Moderate ✓ Your value
0.0 – 0.25	Weak
< 0.0	Poor (clusters overlapping)

Your value (0.4239) means:

- Clusters are **reasonably separated** but not tightly packed ^[1]
- Some points may be **near cluster boundaries** ^[2]
- **Acceptable for real-world data** (most datasets achieve 0.3–0.5)

Davies-Bouldin Score: 1.6507

Range	Quality
< 0.5	Excellent
0.5 – 1.0	Good
1.0 – 2.0	Moderate ✓ Your value
> 2.0	Poor

Your value (1.6507) means:

- Intra-cluster distances are **somewhat large** compared to inter-cluster distances ^[1]
- Clusters have **some overlap or loose boundaries**
- **Not ideal**, but acceptable for complex data

Calinski-Harabasz Score: 122.6928

Quality	Interpretation
Higher = better	Measures ratio of between-cluster vs within-cluster variance ^[1]
Depends on dataset size	Larger datasets naturally have higher scores

Your value (122.6928):

- **Moderate** for most datasets
- Suggests clusters have **reasonable separation** but not optimal compactness
- Comparable datasets typically range 50–500 for moderate clustering ^[1]

Overall Assessment

Quality: Moderate (acceptable but not excellent)

Aspect	Status
Cluster separation	✓ Reasonable (Silhouette 0.42)
Cluster compactness	⚠ Moderate (DB 1.65)
Variance ratio	⚖ Moderate (CH 122.7)

What This Means for Your Pipeline

This pipeline is working but has room for improvement:

Suggestions:

1. Try different DR + clustering combo (e.g., UMAP + HDBSCAN should outperform PCA-based)
2. Adjust hyperparameters:
 - For KMeans: try different `n_clusters`
 - For HDBSCAN: adjust `min_cluster_size`, `metric`
 - For UMAP: tune `n_neighbors`, `min_dist`
3. Check if removing outliers improves metrics
4. Compare with other 5 pipelines to see if this is the best or worst

How to Present in Report

```
\begin{table}
\caption{Clustering Quality Metrics for Pipeline X}
\begin{tabular}{lcc}
\toprule
Metric & Value & Interpretation \\
\midrule
Silhouette Score & 0.4239 & Moderate clustering quality \\
Davies-Bouldin Index & 1.6507 & Moderate (lower is better) \\
Calinski-Harabasz Index & 122.6928 & Moderate (higher is better) \\
\bottomrule
\end{tabular}
\end{table}
```

Conclusion: "The pipeline achieves moderate clustering quality with Silhouette Score of 0.42, indicating reasonably separated clusters. However, the Davies-Bouldin Index of 1.65 suggests some intra-cluster variance that could be improved with alternative DR-clustering combinations (e.g., UMAP + HDBSCAN)."

This is **acceptable for a formal report** — it shows your method works, while acknowledging room for improvement!

✱

1. https://www.linkedin.com/posts/omayma-chattat_clustering-machinelearning-datascience-activity-7244593977306267648-yadI
2. <https://medium.com/@haataa/how-to-measure-clustering-performances-when-there-are-no-ground-truth-db027e9a871c>

